

# Scope

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Scope is the region of codes in which an identifier declaration is active  
global scope-declared outside a function

local scope-declared inside a function

3 kinds: file scope, function scope, block scope

file scope=global scope, can be accessed by any functions in the same

file/other file with proper external declarations

global variables are initialize to 0

all function identifiers have file scope

thus all functions are global in C++

function scope<local scope; all variables/constants declared in the formal  
parameter list or inside the function body have function scope

also called local variables/constants->can only be accessed within the  
function

\*\*\*created when the function is **called**, destructed when the function  
returns\*\*\*

block scope< local scope

a block scope is created when you enclose codes within {}, eg for/while/do-  
while/if/else/switch;

only used within the block

**\*cannot declare a variable/constant of the same name in the same scope**

When an identifier is declared more than once but under different scopes,  
the compiler associates an occurrence of the identifier with its

declaration in the **innermost** enclosing scope.-> 比如在int main开始定义j, 在另  
一个for loop中又定义j, 这时候除了这个for loop中会使用main中j, 但在这个for loop会使  
用for loop中的j